Galaxy Cluster Masses and the Matter Density of the Universe

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This report aims to show that the matter density, Ω_M, is much larger than the baryonic matter density that can be detected. Thus, the majority of matter in the Universe can't be detected. The XCOP galaxy clusters are used to find a value of Ω_M for the Universe. This is done by fitting a Gaussian to the velocity dispersion histograms of the clusters to find a standard deviation, and hence velocity dispersion for each cluster. This is used to find the mass at an overdensity of 200 and Ω_M. Six of the twelve clusters were deemed appropriate to analyse due to their Gaussian velocity distributions and the following Ω_M values were obtained; 0.355 ± 0.044 for A2029; 0.247 ± 0.038 for A2142; 0.257 ± 0.135 for A2319; 0.371 ± 0.079 for A3266; 0.544 ± 0.078 for RXC1825; and 0.340 ± 0.072 for ZW1215. These were in agreement with previous literature [9] and show that $\Omega_M > \Omega_b$, as expected. The impact of the amount of data used on the final result is also examined and the expected inverse proportionality is found. Velocity dispersion as a function of distance from the cluster centre is also considered.

INTRODUCTION

In 1933, F. Zwicky used the virial theorem to determine the velocity dispersion of the Coma cluster and found it to be around 2 orders of magnitude greater than expected from the amount of matter seen in the cluster [1]. These results were replicated by S. Smith in 1936 [1] and again by Zwicky in 1937. In 1970, observations of 21cm radio waves revealed that the rotation curves of M33 and NGC 300 peaked at far larger radii than the detectable mass would predict, suggesting that there is a large density of dark matter far out from galaxy centres [1]. As of today, the search to find dark matter is still ongoing, with the two most promising candidates being axions - low mass bosons - and weakly interacting massive particles (WIMPS) [1].

The identification of cosmological parameters is vital to understanding the evolution of our Universe [9]. Also, cosmological parameters can be used to test theories of gravity[2]. Therefore, this is an important area of research for understanding our Universe.

This paper aims to replicate the findings that $\Omega_b << \Omega_M$ [3] where Ω_b is the fraction of the critical density, ρ_c, due to baryonic matter, and Ω_M is the fraction of the critical density due to total matter. The XCOP clusters will be used, where appropriate, under the assumption that they are large enough to be representative of the Universe [3]. The velocity dispersion, σ_v, will be found from the standard deviation of a Gaussian fitting to the velocity histogram of each cluster.

The virial radius will be approximated with an overdensity of 200, i.e. the radius, R_{200}, which contains a mean density $\rho_{200} = 200\rho_c(z)$. ρ_c was calculated from the Friedmann equation, using constants and their associated errors taken from the WMAP9 cosmology [4]. M_{200} refers to the mass enclosed by R_{200}.

$$M_{200} = \sqrt{\left(\frac{3\sigma_v^2}{G}\right)^3 \frac{3}{4\pi\rho_{200}}} \tag{1}$$

$$R_{200} = \left(\frac{3M_{200}}{4\pi\rho_{200}}\right)^{\frac{1}{3}} \tag{2}$$

Equations 1 and 2 are derived from the virial theorem [5] and the Friedmann equations [6], where G is the gravitational constant. These will be used to calculate R_{200} from σ_v. The procedure to find σ_v will be iterated over until R_{200} from Equations 1 and 2 is equal to the radius searched over in the SIMBAD query. This is done as an accuracy measure to ensure these two values are equal. From here on, this will be referred to as the R_{200} self-consistency procedure.

$$M_{gas} = f_{gas_{Eckert}} \times M_{HSE} \tag{3}$$

$$f_{gas} = \frac{M_{gas}}{M_{200}} \tag{4}$$

To find Ω_M, Equations 3 and 4 will be used to find the mass of gas in the cluster, M_{gas}, and the gas fraction of total cluster mass, f_{gas}. These require using values for $f_{gas_{Eckert}}$ and M_{HSE} from [7] to find values relevant to our data. This works as M_{HSE} is found over the same virial radius as M_{200}.

$$\Omega_M = \frac{\Omega_b}{f_{star} + f_{gas}} \tag{5}$$

Equation 5 will be used to find Ω_M for each cluster. The denominator in Equation 5 is equal to the fraction of baryonic matter, f_b. Here, f_{star} is taken as 0.015 ± 0.005 [7].

EXPERIMENTAL DETAILS

The object type, right ascension, declination, redshift and radial velocity for every object within 20 arcminutes of each of the twelve XCOP clusters were read into a pandas dataframe from SIMBAD using the Astroquery library. The coordinates of each cluster were taken from previous work by Ghirardini et al [8]. Histograms of radial velocity and redshift for each cluster were plotted to decide which data sets were appropriate to analyse using a Gaussian fitting. The initial search radii were taken from Ghirardini et al [8]. A1644 and A85 were discarded for having no clear peak as the data would not be appropriate for the Gaussian fitting method to find σ_v. A1795, A3158, and A2255 were discarded for showing more than one clear peak as this also would not gain an

accurate σ_v from this method as they don't follow a Gaussian shape. A644 was discounted for having too little data as it can be shown from Figure 4 that smaller data sets give less accurate and precise values with this method. A2029, A2142, A2319, A3266, RXC11825, and ZW1215 were included in this analysis as they had clear Gaussian shapes. These will be referred to as the selected clusters from here onwards.

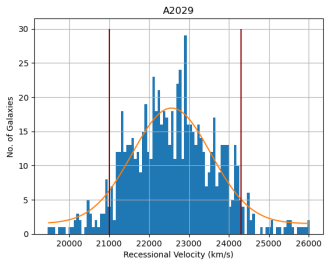


FIG. 1: An example plot of the Gaussian fitting procedure for A2029's radial velocity histogram, the red lines show the 'main cluster' which was used to divide background and main cluster galaxies.

The data frames were filtered to only include objects classed as galaxies. For each cluster, two redshift limits were chosen by eye from its redshift histogram to cut the data set down to just the relevant cluster galaxies, as many of the data sets included small groups of galaxies at much higher redshifts than the main cluster. Figure 1 shows an example of A2029's radial velocity histogram within these limits. Each Gaussian was fitted including a non-zero background value to account for any galaxies outside of the cluster, as otherwise the function would seek to plateau either end at zero. To find this background value, two points were chosen by eye, from radial velocity histograms with both 100 and 2000 bins, which marked roughly the edges of the main cluster. The red lines in Figure 1 show these points as chosen for A2029. The number of data points outside of the main cluster region was divided by the radial velocity range to find a rate of background galaxies per unit radial velocity. This value was multiplied by the bin width of the histogram and the Gaussians were fitted to take this as the zero value of the fit. This was done to reduce the human error from choosing redshift cut off points by eye, as this process renders them arbitrary if the Gaussians are fitted to data in the same redshift range as the background value was calculated with.

The standard deviation from these was taken as the velocity dispersion, σ_v, which was then used to complete the R_{200} self-consistency procedure. After each iteration, the new search radius became the average value of the last search radius and R_{200} as found from Equation 2. R_{200} was accepted as self-consistent when it was less than 0.01 arcminutes from the search radius. To prevent an infinite loop from occurring, this condition was repeatedly increased after various iterations, and the loop was stopped completely after the thousandth iteration. Equation 5 was then used to find Ω_M, with M_{HSE}, f_{star}, and $f_{gas_{Eckert}}$ taken from [7], and $\Omega_b = 0.0463 \pm 0.0009$ taken from [4]. The error in the velocity dispersion was taken from bootstrapping this process over 5000 loops [10]. The errors for all values taken from literature were taken from each paper. The errors for Ω_M were found through Monte Carlo analysis [10].

To see how few radial velocities are needed to produce a precise and accurate value of Ω_M, the A2029 data set was cut down to reduce its length by a factor of $2^{-\frac{n}{5}}$ repeatedly, where n is the current iteration. This factor was chosen as it gave 14 results before the sample was too small to fit a Gaussian to and this plot was found to give the desired detail with 14 data points, and finding any more values greatly increased the time taken to run the code.

The radial velocity of galaxies was plotted against the distance from the cluster centre in arcseconds for each cluster to find if the initial cluster filtering process could be improved by this. The distance from the cluster centre in arcseconds was calculated by SIMBAD using the coordinates entered for each query.

RESULTS

Table I shows the Ω_M values found and their associated errors for the selected clusters. Ω_M is expected to be roughly 0.315 ± 0.007 [9]. Hence, this lies within the error bars of the results from A2029, A2319, A3266, and ZW1215. This suggests the errors were not properly accounted for the remaining two clusters.

Cluster Name	Ω_M
A2029	0.355 ± 0.044
A2142	0.255 ± 0.038
A2319	0.257 ± 0.135
A3266	0.371 ± 0.079
RC1825	0.544 ± 0.078
ZW1215	0.340 ± 0.072

TABLE I: The calculated values of Ω_M for each of the selected clusters.

It can be seen that A2139 has the largest error bar and a percentage error of over 50%. This is like due to the periodicity of the radial velocity data, likely due to low precision measurements. Figure 2 shows that this is also present in the excluded cluster A2255 which suggests that neither cluster should have been included.

Additionally, Ω_M found for RXC1825 is notably larger than the rest of the results, with none of them being within the error bars for RXC1825's Ω_M. Figure 3 shows the RXC1825 data set has a similar distribution to that of the discarded cluster A85 and that this high result is likely a result of having an unclear peak of velocities, leading to a very wide Gaussian fitting.

Figure 4 shows the percentage error on Ω_M as a function of the reciprocal of the amount of radial velocity values used in the analysis. The linear fit had an adjusted R^2 value of 0.7,

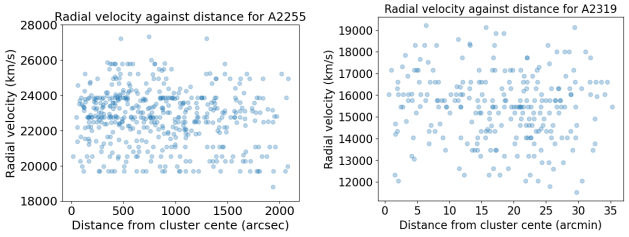


FIG. 2: Radial velocity of each galaxy against its distance from the cluster centre, for clusters A2255 and A2319, respectively.

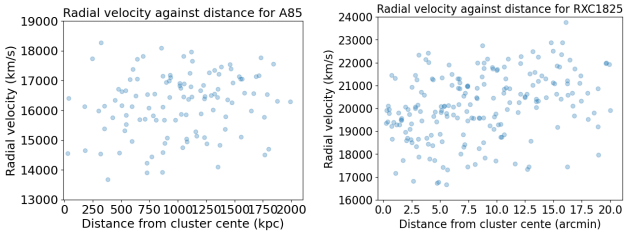


FIG. 3: Radial velocity of each galaxy against its distance from the cluster centre, for clusters A85 and RXC1825.

this indicates that the precision of Ω_M, or more specifically σ_v, is inversely proportional to the amount of data used to fit the Gaussian. Hence, the larger the data set the more precise the result will be.

Figure 5 shows the value of Ω_M as a function of the size of the data set. There is no linear relationship, as the code cut a new random set of data points at each iteration, this shows that the accuracy of results are dependant on the quality of the data, which is often increased as the amount of data increases, thus this result is likely a side effect of this. Also, it can be seen that the results become more consistent with over roughly 400 data points. This is a guideline for the ideal size of a data set however, it should be noted this also depends on the accuracy of the data itself.

Overall, Table I shows $\Omega_b << \Omega_M$ for $\Omega_b = 0.0463 \pm 0.0009$ [4], in line with previous findings [3].

DISCUSSION

The values of Ω_M agree with the previous literature that $\Omega_b << \Omega_M$, suggesting the existence of dark matter [3]. The accepted value of Ω_M [9] is within the error bars of 4 out of 6 of these results. This suggests errors were not properly accounted for in at least two of the clusters.

The error analysis did not take into account errors in the R_{200} value. The self-consistency procedure increases its accuracy, however the difference between the final value and search radius should be accounted for in the error, potentially by taking this difference as the error either side of R_{200}. Also, this procedure could be improved by making the final R_{200} that which had the lowest difference from the search radius so far when the code reaches the maximum amount of iterations.

The Gaussian fitting method is likely the source of error for

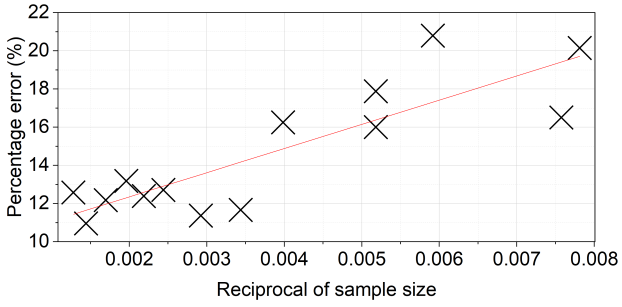


FIG. 4: A plot of the change in percentage error of Ω_M as the size of the data set increases, made by randomly deleting rows from the A2029 data set.

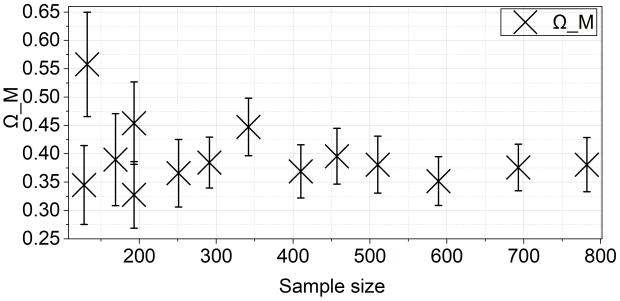


FIG. 5: A plot of the change in Ω_M as the size of the data set decreases, made by randomly deleting rows from the A2029 data set.

cluster RXC1825, as in Figure 3, as well as its radial velocity histogram, show no clear peak density at the cluster centre, and so the fitted Gaussian was likely widened by this, causing a larger velocity dispersion, and hence Ω_M to be found.

Although $\Omega_M = 0.315 \pm 0.007$ from Planck 2018 [9] is not within the error bars of A2142's Ω_M, $\Omega_M = 0.287 \pm 0.009$, the value from the less recent WMAP9 data used [4], does lie within these error bars.

A value of $\Omega_M = 0.287 \pm 0.009$ [4] was assumed while calculating ρ_{200} as the WMAP9 cosmology was used in calculating ρ_c. In future experiments, a self-consistency procedure should be used for this also to reduce errors. Notably, the calculated Ω_M have a smaller average squared difference from the Planck data $\Omega_M = 0.315 \pm 0.007$ than from the WMAP9 data $\Omega_M = 0.287 \pm 0.009$. This is likely because the Planck satellite was used to collect the data used for the coordinates and redshifts of each cluster [8], and may have been involved in the collection of other data used. Also, the more recent Planck Ω_M is considered more accurate due to the higher sensitivity and resolution of the satellite, and as all the data on these clusters was taken after the Planck satellite first launched, it is likely to share a similarly high resolution and sensitivity and hence produce similar results [11].

Error may also have been introduced by selecting cut off points to mark the beginning and end of each cluster by eye. This part of the process should be included in the error prop-

agation by completing a Monte Carlo analysis using 100 random values of the main cluster edges to find an error associated with the background rate. This could then be included in the final Monte Carlo analysis used to find the error associated with Ω_M. The background rates were not varied in the bootstrapping process, despite the data the Gaussian was fitted to changing. This also impacts the validity of the calculated errors.

Deciding cut off point for the cluster would be more effectively done alongside radial velocity against distance from cluster centre plots. This way, those which do not fit the expected caustic shape could be excluded, as shown in Bilton & Pimbblet (2018)[12]. Also, a clustering algorithm, such as Friends of Friends (FoF), could be used to identify cluster members, however this method is sensitive to noise and hyper parameters entered by the user which could introduce more error [18]. Adapted versions of FoF which are less sensitive could be used instead, such as sOPTICS [18].

In Equation 2, a spherical volume is assumed, however in many clusters this is not the case when considering dark matter haloes [16], especially for clusters that have undergone mergers [15], such as A2142 [12]. Thus, using a more accurate shape for each cluster would likely provide more accurate results of Ω_M.

Figures 4 and 5 show the expected inverse proportionality between precision and sample size. However, the adjusted R^2 value of 0.7 suggests 30% of the variation in percentage errors is not explained by sample size [17]. The Gaussian fitting method is dependant on the distribution of the data and so this is likely also a large factor in the precision and accuracy of the result.

If this method were to be used again, the method of choosing which clusters to analyse should be changed as choosing those that appear to have the most Gaussian shaped radial velocity distribution adds human error,. Also Figures 2 and 3, suggest it is inaccurate. It may have been better to calculate Ω_M for all the XCOP clusters as the bootstrap method accounts for errors in distributions, as it did for A2319 [10]. Additionally, a goodness of fit test could be used to determine how well each distribution fits that of a Gaussian [17].

Despite that the radial velocities of galaxies in a cluster would be expected to have a Gaussian distribution due to missing or inaccurate data [13], data does not always show this. Therefore, it may be more appropriate to use a method of finding the velocity dispersion which is not reliant on the distribution of the data. Instead, the biweight estimation may be more appropriate for real data sets [14].

CONCLUSIONS

The calculated Ω_M values successfully demonstrate $\Omega_b << \Omega_M$, in line with previous literature. Five out of the six clusters analysed gave an Ω_M with error bars covering either or both of the values found from WMAP9 data and Planck 2018 data [9][4], which is expected as the WMAP9 cosmology was used in the calculations and some data came from the Planck satellite [8]. The amount of data is inversely proportional to the precision of Ω_M, however it is clear many other factors such as data precision also affect this. Future research seeking to replicate these results would be recommend to use a method of finding the velocity dispersion which is not reliant on the distribution of radial velocity data, and to be more thorough in error propagation. Also, it is noted that plotting the radial velocity against distance from cluster centre is useful for determining cluster limits.

REFERENCES

[1] *History of dark matter*, G. Bertone et al, Reviews of Modern Physics **90**, (2018).

[2] *Cluster gas fraction as a test of gravity*, B. Li, et al, Monthly Notices of the Royal Astronomical Society, **456**, 146, (2016).

[3] *The baryon content of galaxy clusters: a challenge to cosmological orthodoxy*, S. White et al, Nature, **366** 429. (1993).

[4] *Nine-year Wilkinson Microwave Anisotropy Probe (WMAP) Observations: Cosmological Parameter Results*, G. Hinshaw, The Astrophysical Journal Supplement Series **208**, (2013).

[5] *Virial Theorem of Clausius From the Quantum World to Cosmological Systems.*, Vishwamittar, Indian Academy of Sciences **28**, 1719, (2024).

[6] *Review of Particle Physics*, S. Navas et al, Physical Review D **110** 433, (2024).

[7] *Non-thermal pressure support in X-COP galaxy clusters*, D. Eckert, Astronomy and Astrophysics **621**, 2070 (2019).

[8] *Universal thermodynamic properties of the intracluster medium over two decades in radius in the X-COP sample*, V. Ghirardini et al., Astronomy and Astrophysics **621**, A411, (2019).

[9] *Planck 2018 results. VI. Cosmological parameters*, N. Aghanim et al, A & A. **641**, A6 (2020), [arXiv:1807.06209].

[10] *Error estimation in astronomy: A guide*, R. Andrae, arXiv e-prints, **arXiv:1009.2755**, (2010).

[11] *Planck 2018 results. I. Overview and the cosmological legacy of Planck*, Planck Collaboration, Astronomy and Astrophysics, **641**, (2020).

[12] *The kinematics of cluster galaxies via velocity dispersion profiles*, L. E. Bilton et al, Monthly Notices of the Royal Astronomical Society **481**, 1507–1521, (2018).

[13] *Velocity distributions in clusters of galaxies*, A. Faltenbacher et al, Monthly Notices of the Royal Astronomical Society **369**, 1698, (2006).

[14] *Measures of Location and Scale for Velocities in Clusters of Galaxies—A Robust Approach*, T. C. Beers, K. Flynn, and K. Gebhardt, The Astronomical Journal **100** 32, (1990).

[15] *The Galaxy Cluster Mass Scale and Its Impact on Cosmological Constraints from the Cluster Population*, G. W. Pratt et al., Space Science Reviews **215**, (2019).

[16] *The intracluster gas fraction in X-ray clusters: constraints on the clustered mass density*, A. E. Evrard, Monthly Notices of the Royal Astronomical Society **292**, (1997).

[17] *Goodness of Fit in Regression Analysis – R 2 and G 2 Reconsidered*, C. Hagquist et al. Quality & Quantity **32**, 229–245 (1998).

[18] *sOPTICS: a modified density-based algorithm for identifying galaxy groups/clusters and brightest cluster galaxies*, Hai-Xia Ma et al, Monthly Notices of the Royal Astronomical Society **537**,1504-1517, (2025).